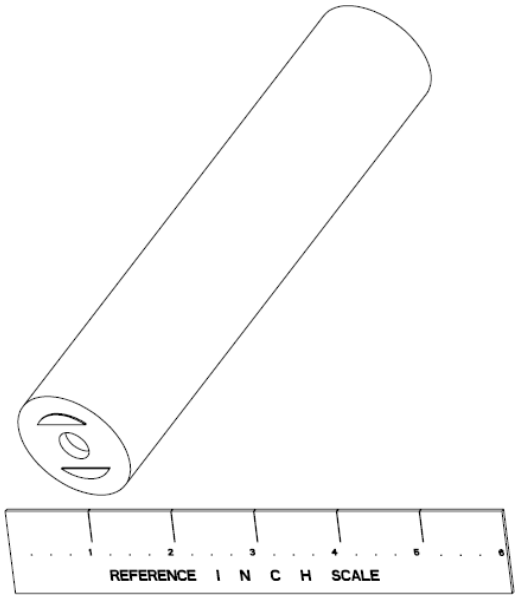
INCH-POUND
MIL-PRF-32271/5A
03 JUNE 2025
SUPERCEDING
MIL-PRF-32271/5
30 OCTOBER 2008

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-32271.



See NOTES for nominal dimensions; see applicable drawing for dimensional requirements.

FIGURE 1. General view, M32271/5 battery shape

For clarity, MIL-PRF-32271 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32271/5A tables are labeled with lower case Roman numerals (e.g. ix).

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type	I
Class	2
Features	C
PIN	M32271/5-12C

Specification requirements: The following requirements of MIL-PRF-32271, identified therein as "when specified", are applicable as indicated below:

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TABLE i. *"When specified" requirements*

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/5-)
			12C
Parallel discharges	*Tables II, III, IV, IX, X, XI, & XII*		*No*
Additional material requirements	*3.3.5*	*4.6.1.11*	*No*
Parallel cell arrangements	*3.4*		*No*
Connectors	*3.4.5*	*4.6.1.7*	*No*
Battery charger connection	*3.4.5.1*	*4.6.1.7.1*	*No*
Connection integrity	*3.4.5.2*		
Static	*3.4.5.2*	*4.6.1.7.2*	*No*
Dynamic	*3.4.5.2*	*4.6.1.7.3*	*No*
Terminal integrity	3.4.6		
Socket Strength	*3.4.6.1*	*4.6.1.8.1*	*No*
Terminal Strength	3.4.6.2	4.6.1.8.2	Yes
Complete discharge device	3.4.7	4.6.1.9	Yes
State of charge device	*3.4.8*	*4.6.1.10*	*No*
State of charge data output	*3.4.8e*	*4.6.1.10*	*No*
Additional design and construction requirements	*3.4.11*	*4.6.1.12*	*No*
Cell charging	3.5.3	4.6.2.3	Yes
Nail penetration	3.5.4	4.6.2.4	Yes
Parallel string charge protection	*3.5.6*	*4.6.2.6*	*No*
Cell series string short circuit	3.5.7	4.6.2.7	Yes
Battery charge protection	3.5.12	4.6.2.12	Yes
Over-current protection	3.5.14.1	4.6.2.14	Yes
Over-temperature protection	3.5.15	4.6.2.15	Yes
Projectile	3.5.16	4.6.2.17	Yes
Additional safety requirements	*3.5.18*	*4.6.2.18*	*No*
Capacity LR	*3.6*	*4.6.4.1.5*	*No*
Capacity LRT	*3.6*	*4.6.4.1.9*	*No*
Capacity IDC	*3.6*	*4.6.4.1.10*	*No*
Surface temperature	3.6h	4.6.4.1	Yes
Immersion	3.7.8	4.6.5.8	Yes
Watertight integrity	*3.7.9*		
Shallow	*3.7.9*	*4.6.5.9*	*No*
Deep	*3.7.9*	*4.6.5.10*	*No*
Electromagnetic interference	*3.7.10*	*4.6.5.11*	*No*
Electrostatic discharge	*3.7.11*	*4.6.5.12*	*No*
Additional environmental requirements	*3.7.12*	*4.6.5.13*	*No*

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TABLE i. *"When specified" requirements* – Continued

Requirement	Requirement Paragraph	Verification Paragraph	Applicability by PIN Suffix (M32271/5-)
			12C
DO NOT CHARGE! marking	*3.8.1.2b*		*No*
Complete discharge device marking/label	3.8.4		Yes
State of charge marking	*3.8.5*		*No*
Simplified battery label	3.8.7		Yes
Additional markings	*3.8.8*		*No*

Dimensions, marking and configuration

Battery - Drawing A3315882
Battery Connector – Not Applicable
Mating Connector – Not Applicable
Battery Charger Connector – Not Applicable
Simplified Battery Label (3.8.7) – The energy shall be calculated using the I test data.

Battery voltages (3.4.4 & 4.6.1.5):

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):

Maximum – 9.9 volts
Minimum – 9.0 volts

Battery closed circuit voltage (3.4.4.2 & 4.6.1.5.2):

For capacity, initial voltage delay, and closed circuit voltage test:
Minimum – 6.0 volts

Cell closed circuit voltage test (3.4.4.3 & 4.6.1.3):

Minimum - 2.0 volts per cell

Maximum weight (3.1 & 4.6.1.6):

1.2 lbs. (544 g)

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Capacity test requirements (3.6 & 4.6.4):

Test	Initial voltage delay Time (MAX) to Volts		Minimum Capacity		Cut-off Volts
			Storage Period		
			1-Week	4-Week	
I	1 second	6.0	5.00 hours		6.0
L	1 second	6.0	2.75 hours		6.0
H	1 second	6.0	8.30 hours		6.0
IT	1 second	6.0	4.7 hours	4.5 hours	6.0
LT	1 second	6.0	2.6 hours	2.5 hours	6.0
HT	1 second	6.0	7.7 hours	7.5 hours	6.0

Abuse test pulse discharge minimum capacity requirement (3.5.13 & 4.6.2.13l):

9.4 ampere- hours

METHODS OF EXAMINATION AND TEST:

Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell with 2 amperes constant current, 1 ohm constant resistance, or 4 watts constant power.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load each battery with 2 amperes constant current, 3 ohms resistance, or 12 watts of power.

Abuse test, pre-discharge: When tested as specified in 4.6.2.13a, discharge with a load of 2.0 amperes for 2.5 hours.

Abuse test, pulse discharge: When tested as specified in 4.6.2.13l, discharge with a load of 2.5 amperes for 1 minute followed by 1.8 amperes for 4 minutes, cycled continuously to 6.0 volts.

Battery over-current protection: When tested as specified in 4.6.2.14, load batteries with either 4 amperes or 20 watts.

Immersion: When tested as specified in 4.6.5.8, apply a load of 1.2 amperes during the storage and immersion time periods specified.

Capacity tests (4.6.4):

Test	Discharge Rate	Duty Cycle
I	2.0 amperes	Continuous discharge to cut-off volts, followed by 2.0 amperes To zero volts, then 2.0 amperes forced discharge for 5 minutes.
L	2.0 amperes	Continuous discharge to cut-off voltage
H	2.0 amperes	Discharge for 5 minutes at 2 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage
IT	2.0 amperes	Continuous discharge to cut-off voltage
LT	2.0 amperes	Continuous discharge to cut-off voltage
HT	2.0 amperes	Discharge for 5 minutes at 2 amperes, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage

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NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This specification covers non-rechargeable lithium batteries of the non-reserve type composed of electrochemical cells using lithium metal in the anode. They are intended for use in military communications and electronics equipment.

Nominal ratings[1]: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

Battery PIN:	**M32271/5-12C (Li/MnO_2)**
Overall Dimensions::	7.32 in. L x 1.55 in. DIA
MAX Weight:	1.2 lbs (544 grams)
Voltage Range:	6.0-9.9
Nominal Capacity: (in ampere-hours)	10 Amp-hrs
Nominal Energy: (in watt-hours)	78 Watt-hrs
MAX rated power output:	23 watts
MAX continuous load rating:	2.0 amperes
MAX pulse load rating:	2.5 amperes
Instantaneous trip rating:	4.0 amperes
Operating temperature range:	-4 to 130ºF (-20 to 55ºC)
Storage temperature range:	-40 to 160ºF (-40 to 71ºC)
MAX abusive temperature: (non-operating)	195ºF (91ºC)

Other data:

Military Type Designations. The military type designation that relates to the PIN covered by this specification sheet is as follows: The BA-5360()/U designation has been used for the Type I PIN. It should be noted that the former Type II BA-5600()/U designation has been replaced by the BA-5360()/U.

This battery is subject to the transportation requirements of 49 CFR 173.185. The battery has solid cathodes and more than 2.0 grams of lithium content. See applicable Material Safety Data Sheet (MSDS) for the maximum lithium weight per cell and battery. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

This battery includes a complete discharge device in order to render the battery non-reactive after use. If the battery is damaged, or if the device does not operate as indicated by an amber or yellow light, the battery is considered reactive waste. This battery may be recycled after use. Universal waste rules, where applicable, apply only when recycling is the chosen disposal method.

[1] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): https://www.battery.army.mil.

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The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using the Type I battery covered by this specification sheet: Technical Bulletin 43-6135. Please contact the CECOM LCMC Directorate for Safety at usarmy.APG.cecom.mbx.amsel-sf-sec-cecom-lcmc--ds@army.mil for further information.

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Preparing activity:
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